

EEG-Based Neurological Outcome prediction in Acute Coma and/or Comatose Survivors of Cardiac Arrest Patients using LSTM Model

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Abstract Cardiac arrest carries an in-hospital mortality of 75%, with 20–40% of survivors developing significant neurological disability. Early prognostication of neurological outcomes in post-anoxic coma is critical for guiding decisions regarding life-sustaining therapies, yet current approaches suffer from high inter-observer variability and limited sensitivity in the acute phase. To develop and validate a deep learning framework for automated, objective prediction of neurological outcomes in comatose cardiac arrest patients using continuous EEG recordings. We analyzed EEG recordings from the I-CARE v2.1 database. Signals were preprocessed and a hybrid CNN–LSTM architecture was trained to capture spatial and temporal dynamics. Both standard 10–20 (19 channels) and longitudinal bipolar (16 channels) montages were evaluated. Performance was assessed via AUC-ROC through a CNN–LSTM that achieved maximum AUC 0.88 at 32–36–40–44 hour range post-arrest using the standard montage. The CNN–LSTM framework provides robust, automated prognostication from post-cardiac arrest EEG, with clinically meaningful discrimination superior to conventional machine learning approaches. Integration into critical care workflows may reduce prognostic uncertainty and support objective, timely decision-making for life-sustaining therapy.

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1 Introduction

Cardiac arrest constitutes a medical emergency with an approximate in-hospital mortality rate of 75%, and nearly 20–40% of survivors develop significant neurological disability that limits long-term functional independence [1, 2]. This condition involves an abrupt interruption of oxygenated cerebral blood flow, leading to hypoxic-ischemic injury that compromises cortical neuron viability and often results in post-anoxic coma [3, 4]. These injuries are associated with severe functional impairments, including cognitive, sensory, and motor deficits. Therefore, a central clinical challenge is accurately assessing the extent of neurological damage. In particular, the prognostic assessments play a critical role in guiding clinical decisions, including the continuation or withdrawal of life-sustaining therapies [5].

International guidelines emphasize early and accurate prognostication after return of spontaneous circulation and recommend a multimodal approach, combining clinical, electrophysiological, and imaging data [6, 7]. Nevertheless, prognostic uncertainty remains high in the early post-arrest period due to the limited sensitivity of the used modalities and substantial inter-patient variability. Electroencephalography (EEG) provides a noninvasive means of continuously monitoring brain activity and has been widely used to assess cerebral dysfunction in these patients [3]. However, its clinical utility is limited by observer variability and the complexity of long-duration recordings. Moreover, the translation of qualitative EEG patterns into consistent clinical decisions often lacks standardized criteria, motivating the development of quantitative approaches that extract objective, reproducible signal features [8, 9].

Most quantitative EEG characterization methods rely on predefined biomarkers and handcrafted features, such as spectral power or entropy, which are used in statistical or machine learning models for prognostic assessment [10]. However, these approaches typically treat EEG as static statistical summaries, potentially ignoring the temporal dynamics of brain activity during recovery.

In this work, we propose a data-driven approach based on Long Short-Term Memory (LSTM) networks to characterize EEG signals in comatose patients following cardiac arrest. The method explicitly learns temporal dependencies in post-arrest recordings, enabling the identification of dynamic patterns associated with neurological recovery and providing a framework that extends static EEG representations.

2 Materials and pre-processing

2.1 Data

The study uses the I-CARE v2.1 database hosted on PhysioNet [14], which includes 607 EEG recordings from comatose patients collected across multiple hospitals in the United States and Europe. The dataset is heterogeneous in terms of acquisition protocols, including variability in channel numbers, sampling frequencies, and elec-

trode configurations across institutions. Recordings typically include 19–21 EEG channels and additional physiological signals, with monitoring durations averaging approximately 65 hours but showing substantial variability across patients. The cohort also exhibits demographic and outcome variability, reflecting real-world clinical conditions and posing challenges for robust, generalizable modeling. The input variables consist of EEG signals, while the output variable corresponds to the CPC outcome (Good or Poor). The class distribution is imbalanced, with 62% of cases labeled as Poor and 38% as Good..

2.2 Pre-processing

EEG signals were preprocessed following standard clinical and signal-processing guidelines. First, the data were band-pass filtered between 0.5 and 45 Hz using a zero-phase FIR filter implemented in MNE-Python, and subsequently resampled to 128 Hz in accordance with the Nyquist-Shannon sampling theorem, ensuring that the sampling frequency satisfied $f_s \geq 2f_{max}$ and $f_{max} = 45Hz$ [12]. Line noise was attenuated using a multi-taper approach without relying on fixed filtering assumptions. A robust average reference was then iteratively estimated while excluding noisy channels, which were identified based on statistical and spectral criteria. Finally, RANSAC was applied to detect channels that deviated from expected signals using spatial predictions from neighboring electrodes [15, 1].

To train the model, EEG signals were analyzed using the standard 10–20 system with 19 monopolar channels. In addition, a longitudinal bipolar montage (anterior–posterior chains) comprising 16 channels was employed, as it enhances the detection of localized brain activity by referencing neighboring electrodes. The reduced number of channels in the bipolar configuration also facilitates the identification of spatial gradients. Both monopolar and bipolar representations were included to provide complementary information. The specific channel configurations follow standard EEG montage definitions as described in [3].

2.3 LSTM (Long Short Term Memory)

A hybrid Convolutional Neural Network-LSTM (CNN–LSTM) architecture was used for spatiotemporal signal processing. This model combines 2D convolutional feature extraction (BatchNorm + ReLU) with a two-layer LSTM (hidden size = 64, dropout = 0.2) and a fully connected classifier for binary prediction. The CNN–LSTM architecture is well-suited for EEG in post-cardiac arrest patients because it captures both spatial and temporal structure. The CNN aims to extract spatial patterns across electrodes (e.g., rhythms or pathological waveforms), while the LSTM models their temporal evolution, which is critical for prognosis. This com-

bination enables effective spatiotemporal representation for reliable binary outcome prediction[13].

2.4 Experimental setup

To model temporal dynamics, EEG data were segmented into time intervals across different hours. Specifically, 3-minute segments were extracted between the 3rd and 5th minutes from four distinct hourly recordings, which were nearly continuous and separated by approximately four hours. Table 1 summarizes the selected time points, covering a range from 24 to 48 hours.

The dataset was divided into two subsets: 75% for training (412 patients) and 25% for validation (138 patients). The model was trained using a stratified 5-fold cross-validation strategy applied to the training set to ensure robust performance estimation. A total of 607 patients were available in the original dataset; however, not all patients had recordings corresponding to the reference hours defined in this study, and therefore only a subset was included in the analysis. Importantly, data partitioning was performed at the patient level to prevent data leakage, even though multiple hourly recordings were available for each patient.

Monitoring windows were defined between 24 and 48 hours after admission, using a 4-hour step (24, 28, 32, 36, 40, 44, and 48 hours), following clinical evidence [11]. For each patient and each monitoring window, a single 180-second EEG segment was extracted between minutes 2 and 5 to reduce initial signal noise. No overlap between segments was considered. The use of short segments was motivated by data availability and computational constraints. Furthermore, the data were segmented according to channel configuration, distinguishing between standard channels and differential time-point channels derived from a longitudinal bipolar montage.

Having a maximum range of 4 hours of data, each segment of 3 minutes in each hour was segmented by a window of 200 timestamps with overlapping of 64 timestamps creating a tensor $x_h \in \mathbb{R}^{N_p \times 19 \times 200}$ where N_p are the quantity of patches with shape of 19 channels and a window of 200 timestamps, h represents the hour taken for each patient.

The model was optimized to maximize discrimination, as measured by the ROC AUC, and the Youden index [16]. This optimization strategy emphasized clinically relevant operating points, particularly those achieving balanced predictive performance across both classes.

Table 1, Group 1 corresponds to standard channels, while Group 2 represents differential time-point channels.

- Group 1: Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, and Pz.
- Group 2: Fp1-F3, F3-C3, C3-P3, P3-O1, Fp2-F4, F4-C4, C4-P4, P4-O2, F7-T3, T3-T5, T5-O1, F8-T4, T4-T6, T6-O2, Fz-Cz, Cz-Pz

Model training was conducted on a personal computer with 32 GB RAM and an NVIDIA GeForce RTX 3080 Laptop GPU (12 GB VRAM). Results were obtained through multiple experiments across different monitoring windows and segmentation strategies. Clinical outcome was defined according to the Cerebral Performance Category (CPC) scale, with CPC scores of 1–2 considered as **good** neurological outcome and scores of 3–5 as **poor** outcome.

3 Results and discussion

Table 1 reports the LSTM model performance, the metrics to evaluate the model are the area under the curve (AUC) and the Youden index given by

$$I_{Youden} = \operatorname{argmax}(Tpr - Fpr)$$

where Tpr and Fpr represent the **True-positive-rate** and **False-positive-rate** respectively, because the accuracy can be misleading (especially with class imbalance), moreover Youden directly optimizes the ROC trade-off allowing choose the best model [16]. As observed, performance varied considerably across selected time intervals, with an recall ranging from 0.63 to 0.94, an specificity ranging from 0.38 to 0.94 being few unstable; the F1-score is ranged from 0.74 to 0.83, AUC values between 0.83 and 0.88. It can see that the channels of both montage have an interesting performance, even at same range hours, among the 32 and and 44 hours the model have presented a high AUC.

Table 1 Performance of the LSTM model across different monitoring windows.

Monitoring hour Group input (h)		Recall	Specificity	F1-score	AUC
24–28–32–36	1	0.74	0.83	0.79	0.85
28–32–36–40	1	0.78	0.78	0.81	0.84
30–34–38–42	1	0.8	0.83	0.83	0.86
32–36–40–44	1	0.7	0.94	0.79	0.88
34–38–42–46	1	0.73	0.89	0.81	0.87
24–28–30–32	2	0.81	0.78	0.82	0.85
28–32–36–40	2	0.78	0.75	0.77	0.79
30–34–38–42	2	0.63	0.89	0.74	0.84
32–36–40–44	2	0.73	0.87	0.80	0.87
34–38–42–46	2	0.94	0.38	0.79	0.83

A set of baseline models, including Logistic Regression, Random Forest, and Support Vector Machine, was trained using handcrafted features such as mean frequency per channel, signal mean, and signal variance, among others. The average performance achieved by these models was approximately 0.59 in terms of AUC. In

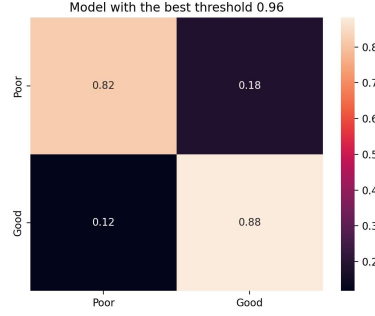


Fig. 1 The matrix confusion above represent the best model in the range hour 32–36–40–44 hour of group 1 montange with a AUC of 0.88 using the best threshold.

contrast, the best-performing model in this study reached an AUC of 0.88, demonstrating a substantial improvement in CPC prediction.

In Figure 1, the confusion matrix illustrates the model’s performance using the optimal Youden index of 0.70 (shown in the right panel). It can be observed that the predictions are relatively well balanced along the diagonal of the matrix, indicating good classification performance for both the “Poor” and “Good” outcome classes using a threshold of 0.96. The best-performing configuration corresponds to the time window range of 32, 36, 40, and 44 hours (Group 1), likely reflecting a reduced influence of transient physiological effects, sedation, or signal noise during the acute phase.

Taken together, these findings suggest that longer, temporally contiguous EEG segments recorded during intermediate monitoring periods capture more stable and discriminative patterns, thereby enhancing predictive performance. From a methodological standpoint, AUC consistently proved to be a more robust evaluation metric than accuracy, as it better reflects model discrimination in the presence of class imbalance.

Overall, the results confirm that the proposed LSTM-based framework effectively exploits temporal dependencies inherent in EEG signals. Careful selection of monitoring windows, segmentation strategies, and model configuration is essential to optimizing predictive performance while maintaining the computational efficiency required in clinical settings.

4 Conclusion

The proposed CNN–LSTM framework achieves robust, clinically meaningful discrimination (AUC = 0.88) for neurological outcome prediction in comatose cardiac arrest patients, substantially outperforming conventional machine learning baselines (mean AUC = 0.59). Optimal predictive performance is obtained during intermedi-

ate post-arrest monitoring windows (32–36–40–44 hour range), suggesting that EEG patterns in this period capture more stable signatures of hypoxic-ischemic brain injury once transient physiological effects and sedation have diminished. The use of patient-level stratified cross-validation ensures generalizable results and prevents data leakage, addressing a critical methodological requirement for clinical translation. These findings support the integration of deep learning-based EEG analysis into multimodal prognostication pathways as an objective, reproducible decision-support tool. Future work should validate this approach in prospective cohorts and explore real-time implementation to reduce inter-observer variability and inform timely, evidence-based decisions regarding life-sustaining therapies in critical care settings.

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